Math 55 worksheet Wednesday 3/4/2026

1. What is the greatest number of edges a graph with n vertices can have? Prove that any graph with this many edges has only one connected component.

Solution: In such a graph, every vertex is connected to every other, so each vertex has degree $n-1$. Therefore there are a total of $(1/2)\sum_{i=1}^{n}(n-1) = n(n-1)/2$ edges, by the handshaking theorem.

If any graph has more than one connected component, it cannot be the graph with the most edges, since you can always add an edge between the two components.

2. Let G be a connected graph where all its vertices have degree 2, except for k of them, all of which have degree 1. What are the possible values of k?

Solution: Since almost all the vertices of G have degree 2, the only possibilities are for them to be arranged in a circle ($k = 0$) or in a line ($k = 2$).

3. Let K_6 be the *complete graph on six vertices*, that is to say, the graph with six vertices with every pair of vertices connected by an edge. (In question 1, you determined how many edges this graph has, but you don't need that information for this problem.) Assume we color every edge either red or blue. Prove that, no matter how we do this, there exists a triangle (a set of three vertices, all connected by edges) with all the edges one color.

Solution: Remove the red edges, keep the blue ones, and apply Ramsey theory.

4. Find an example of a graph with five vertices that contains neither a clique with three vertices nor an independent set with three vertices. In other words, prove that Ramsey's theorem, as discussed in class, needs six vertices to work.

Solution: The cycle graph with five vertices will do it: there's no triangle, so no three-vertex clique, and any set of three vertices must contain two adjacent ones, which are connected.

5. A *cycle* is a path that starts and ends at the same vertex, does not contain any other vertex more than once, and contains more than one vertex. Let G be a graph. Assume that there exist two vertices v, w of G, and assume that there are two different paths from v to w. Prove that G contains a cycle. *Hint:* It might be helpful to draw a diagram.

Solution: Following the paths from v, let x be the last vertex before they differ for the first time, and y be the first vertex they both have after x. Then there exist two paths from x to y that don't share any other vertices. Run the first one forward and the second one backward to get your cycle.